Kian (Mohammad) Kianpisheh

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Summary: I am a PhD candidate in Computer Science at the University of Toronto, working under the supervision of Prof. Khai Truong in the Dynamic Graphics Project (DGP) lab. My research lies at the intersection of ubiquitous computing and applied machine learning , focusing on the development of personalized and customizable smart assistants for real-world applications. I design and develop intelligent systems that empower end-users to customize the system behaviour based on their individual habits, preferences, and environment.

EDUCATION

University of Toronto Sep 2018-Present
Ph.D. candidate in Computer Science, DGP lab
Personalized Ubiquitous Intelligent Systems
Advisor: Prof. Khai Truong

Sharif University of Technology 2015-2017
M.Sc. in Electrical Engineering, (Computer Vision, Scene Understanding)
Advisor: Prof. Mohammad Sharifkhani

Shahed University 2008-2013
B.Sc. in Electrical Engineering, (Image Processing)
Advisor: Prof. Alireza Behrad

EXPERIENCE

University of Toronto 2018-2025
Research Assistant — Applied AI
- Designed personalized intelligent systems using multimodal machine learning.
- Worked with motion, vision, audio, and other sensor data to model human behavior and context.
- Focused on adaptive and user-aware AI/IoT applications, combining signal processing with deep learning approaches.

Mathsronauts 2022-2023
Machine Learning Engineer (PyTorch, OpenCV)
- Developed a platform that allowed students to interactively learn and experiment with computer vision and machine learning concepts.

Paya Vision 2013-2015
Machine Learning Engineer (TensorFlow, OpenCV, OpenCL, Scikit-Learn)
- Designed and developed a content-based video retrieval system that allows users to retrieve relevant video clips using natural language queries.

University of Toronto-Teaching Assistant
Courses: Image Understanding, Introduction to Computer Vision, Introduction to Machine Learning, Data Structures and Analysis, Programming on the Web (lead TA), and many more.

RESEARCH AND PUBLICATIONS

Kianpisheh M, Mariakakis A, Truong KN. SAHARA: Self-supervised Approach for Human Activity Recognition based on Everyday Audio Events. (under review)

Kianpisheh M, Mariakakis A, Truong KN. exHAR: An Interface for Helping Non-Experts Develop and Debug Knowledge-based Human Activity Recognition Systems. Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies. 2024 Mar 6;8(1):1-30. (link)

Kianpisheh M, Li FM, Truong KN. Face recognition assistant for people with visual impairments. Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies. 2019 Sep 9;3(3):1-24. (link)

Kianpisheh M, Content-based Video Retrieval in Traffic Videos using Latent Dirichlet Allocation Topic Model. 2017 Sep, arxiv. (link)

SELECT LIST OF TECHNICAL PROJECTS

SAHARA: An audio-based self-supervised human activity recognition system. 2025
The system automatically clusters repetitive audio events and characterizes the user's activity based on those detected events.

IMUdio: A smartwatch human activity recognition system based on audio and IMU sensor data. 2024
A lightweight deep learning model that fuses audio and IMU data embeddings to recognize human activities, with inference performed directly on a smartwatch.

exHAR: An Interface for Helping Non-Experts Develop and Debug Human Activity Recognition Systems. 2024
An explainable framework that enables users to personalize sensor-based human activity recognition systems.

SensorLab: A data visualization tool for wearable devices and smartphone IMU sensor data. 2023
A visualization web application that shows the spectrogram of IMU (i.e., accelerometer and gyroscope) sensor data.

ActivityLab: A data visualization tool for wearable devices and smartphone sensors data. 2022
A web-based visualization application that depicts human activity datasets as a chronological sequence of events. Users can program the system in If-This-Then-That format.

Face recognition assistant for people with visual impairments. 2020
A smart wearable camera that automatically captures the faces of people the user interacts with. It filters out low-quality images (poor lighting, extreme head poses, motion blur) to ensure reliable face classification.

CBVR: Content-based Video Retrieval using Latent Dirichlet Allocation Topic Model. 2017
The system detects traffic patterns using Topic Models. A GUI allows users to search for specific patterns efficiently.

TEACHING

University of Toronto

- Foundation of Computer Vision
- Programming on the Web (Lead TA) (4x)
- Introduction to Computer Science (>4x)
- Computational Thinking
- Image Understanding (2x)
- Data Structures and Analysis
- Introduction to Programming (>4x)
- Great Ideas in Computing

SKILLS

Machine Learning: PyTorch TensorFlow Hugging Face OpenCV

Web development: React React Node JavaScript TypeScript

Design: Figma Illustrator

Mobile Development: Android (Java)

REFERENCES

Khai N. Truong (Professor, Department of Computer Science, University of Toronto)

Alex Mariakakis (Asistant Professor, Department of Computer Science, University of Toronto)